

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Practical Lab Exam 2

Practical Lab Exam 2

Class: Class 9

Assessment Type: Practical Lab Exam 2

Total Marks: 20

Duration: 45 minutes

Topics Covered: SQL with Python and Web Development

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Task 1: SQL Database Operations with Python (5 marks)

Write a Python program using SQLite that:

- Creates a database `school.db` with tables:
 - `Students(roll_no INTEGER PRIMARY KEY, name TEXT, class TEXT, section TEXT, admission_date DATE)`
 - `Subjects(sub_id INTEGER PRIMARY KEY, sub_name TEXT, max_marks INTEGER)`
 - `Results(roll_no INTEGER, sub_id INTEGER, marks_obtained INTEGER, FOREIGN KEY references Students and Subjects)`
- Inserts at least 6 students and 4 subjects with result data
- Executes and displays:
 - Students with marks above 80 in any subject
 - Subject-wise average and highest marks
 - Students ranked by total marks
 - Class-wise performance summary (total students, average marks per class)
- Updates one student's marks and displays the before/after comparison

Task 2: HTML and CSS Web Page (5 marks)

Create a complete HTML page `school_portal.html` with an external CSS file `portal.css` that includes:

- A styled header with school name and navigation links (Home, About, Academics, Contact)
- A hero section with a welcome message and call-to-action button
- A table displaying 6 students with columns: Name, Class, Section, Marks, Grade
- The table should have:
 - Alternating row colours (zebra striping)
 - Hover effect on rows
 - Proper borders, spacing, and responsive design
- A contact form with fields: Name, Email, Subject, Message, and a Submit button
- A styled footer with copyright information
- CSS should use Flexbox or Grid for layout

Task 3: Interactive Web Application (5 marks)

Create a web page `todo_app.html` with embedded JavaScript that:

- Allows users to add new tasks with a title and priority (High, Medium, Low)
- Displays tasks in a list with colour-coded priority indicators
- Allows marking tasks as complete (strikethrough effect)
- Allows deleting tasks
- Shows a summary: total tasks, completed tasks, pending tasks

- Uses CSS for attractive styling with smooth transitions/animations
- Stores tasks in an array (in-memory) and re-renders the list on any change

Viva Voce (5 marks)

1. What is the difference between INNER JOIN and LEFT JOIN? Give one example of each. (1 mark)
2. How does CSS specificity work? Which rule takes priority: inline, internal, or external CSS? Explain with an example. (1 mark)
3. What is the purpose of a **Foreign Key** in SQL? How does it maintain data integrity between tables? (1 mark)
4. Explain the difference between WHERE and HAVING in SQL. When would you use each? (1 mark)
5. What is the difference between **responsive design** and **adaptive design** in web development? Give one example of each approach. (1 mark)